

Euclid's Elements — Book I

Proposition 45

Formal Reconstruction — Technical Documentation

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CGJTEAMLAB

Constructing a parallelogram equal in area to a rectilinear figure

1 Introduction

Proposition I.45 is the culmination of the application-of-areas sequence I.42–I.45. Proposition I.42 constructs a parallelogram equal in content to one triangle and having a prescribed angle. Proposition I.44 adds the location constraint that such a parallelogram be applied to a prescribed segment. Proposition I.45 passes from one triangle to a rectilinear figure. Classically, the new idea is finite composition: triangulate the figure, apply the preceding construction to its triangular pieces, and combine the resulting parallelograms into one parallelogram with the prescribed angle.

The current Lean reconstruction follows that compositional idea, but makes its data flow explicit. A triangulated figure is represented by `HilbertTriangulatedFigure Geo`, and its executable scissors content is the sum of the corresponding triangle terms. For a single triangle the proof calls I.42 directly. For a longer list, the induction hypothesis produces a parallelogram for the tail, the next triangle is converted into an oriented adjacent parallelogram, and the two parallelograms are pasted into a larger one. The paste must preserve both the synthetic content and the prescribed angle.

The final theorem remains entirely synthetic. It does not attach a number to a figure and does not use a numerical area formula. Equality of content is represented by

`HilbertScissorsEquicomplementable`

If L is the input list of triangles, the theorem constructs a parallelogram $STUV$ satisfying

$$\angle STU \cong \angle XYZ$$

and

$$\text{Term}(L) \approx \text{Par}(STUV),$$

where $\approx$ denotes the project's equicomplementability relation.

I.45 also resolves an architectural question left deliberately open earlier in the area block. The existing Hilbert layer already contains `HilbertTriangle`, `HilbertTriangulatedFigure`, and relations between triangulations. For the constructive content theorem of I.45, this list-of-triangles representation is sufficient: the proof needs only finite addition of triangle content and an induction step that adjoins one triangle. No new polygon datatype and no numerical area layer are required.

There is, however, an important representation boundary. In the current library,

```
abbrev HilbertTriangulatedFigure
  (Geo : Geometry.Geo) :=
  List (HilbertTriangle Geo)
```

is only a finite list of triangles. The statement of `euclid_proposition_45` assumes that the list is nonempty and that each triangle is nondegenerate. It does not additionally assert that the triangles have disjoint interiors, fit along common edges, or form the interior of one simple polygon. Thus the formal theorem is a valid and in one sense stronger additive statement about any finite nonempty list of nondegenerate triangles. To read such a list as one geometric rectilinear figure requires a representation invariant supplied outside this theorem. The documentation therefore distinguishes the proved scissors statement from a full internal semantics of polygonal regions.

As in the preceding content propositions, the proof has three distinct layers.

- **Geometry:** extensions, SameRay orientation, SameSide/OppositeSide separation, parallel uniqueness, Pasch, strict betweenness, and recognition of the pasted outer parallelogram.
- **Content:** scissors refinement, equicomplementability of the new triangular piece with an adjacent parallelogram, addition, and transitivity.
- **Representation:** a rectilinear input is represented by a triangle list and a multiset term; the I.44 construction is locally strengthened to retain one strict order fact that its public wrapper intentionally hides.

The most important foundational result is negative: I.45 introduces no new geometric axiom, no new content axiom, no continuity principle, and no positivity or cancellation principle. The temporary `i45_extend_parallelogram` assumption used during development has been completely eliminated and replaced by a theorem.

2 The Classical Configuration

Euclid states the proposition for a given rectilinear figure and a given rectilinear angle. In the first nontrivial case, let the figure be a quadrilateral $ABCD$ divided by the diagonal BD into the two triangles ABD and DBC .

First construct, by I.42, a parallelogram equal to triangle ABD in the given angle. Then apply, by I.44, a second parallelogram equal to triangle DBC to a side of the first parallelogram. The second parallelogram is placed so that the two pieces are adjacent rather than overlapping. Euclid then uses parallel-line angle relations and straight-line criteria to show that the outer boundary is again a parallelogram. Since the two inner parallelograms are respectively equal to the two triangles, their union is equal to the whole rectilinear figure.

For a rectilinear figure with more triangular pieces, the same operation is repeated. One may therefore read the classical construction recursively: start with a parallelogram for one triangle and append one triangle at a time by applying an adjacent parallelogram to the current boundary.

Figure 1 separates the historical two-triangle construction from the later formal normalization. The left panel shows a representative quadrilateral divided by a diagonal; the right panel shows the classical I.42/I.44 operation of replacing the two triangular contents by two adjacent parallelograms whose outer boundary is again a parallelogram.

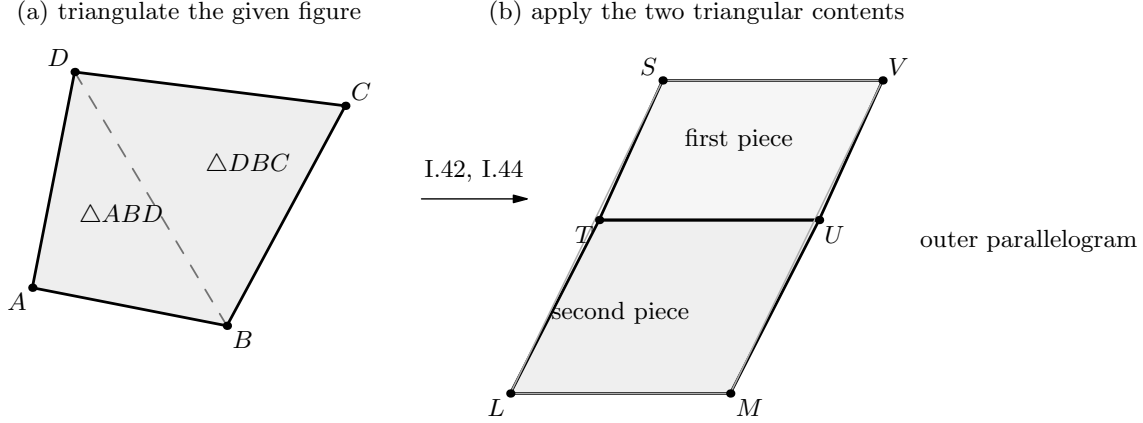


Figure 1: The first nontrivial classical case of I.45. A rectilinear quadrilateral is triangulated by a diagonal. I.42 constructs a parallelogram equal to the first triangle in the prescribed angle, and I.44 applies a second parallelogram equal to the second triangle along the current boundary. The two applied pieces form one larger parallelogram.

The current Lean proof adopts exactly this recursive shape, but its local paste configuration is normalized to the notation

$$STUV, \quad TUML, \quad SLMV.$$

Here $STUV$ is the already constructed parallelogram, $TUML$ represents the next triangle, and $SLMV$ is the pasted outer parallelogram. The strict order conditions are

$$S - T - L, \quad V - U - M.$$

These relations, which are proved rather than read from a drawing, are the formal content of saying that the second piece is attached externally along TU . The corresponding formal geometry is displayed only after those order facts have been constructed in Figure 2.

3 Classical Proof

The classical proof is a finite composition of I.42 and I.44.

For a quadrilateral $ABCD$, draw the diagonal BD . Proposition I.42 gives a parallelogram equal to triangle ABD in the prescribed angle. Proposition I.44 then applies to one side of that parallelogram a second parallelogram equal to triangle DBC , in the supplementary orientation required for the two pieces to form a straight outer boundary.

The angle equalities supplied by the two constructions, together with I.29, show that the relevant sides of the two parallelograms are parallel. Euclid uses I.14 to recognize the straight-line relations created at the common side. The outer opposite sides are then shown parallel using the

established parallel theory, including I.30 and the parallelogram criterion behind I.33. Thus the two applied pieces form one larger parallelogram.

The content conclusion is purely additive. One inner parallelogram is equal to triangle ABD , the other to triangle DBC , and the two triangles are the whole rectilineal figure. Therefore the outer parallelogram is equal to the given rectilineal figure. The prescribed outer angle is inherited from the first construction.

The two-triangle historical DAG is therefore

$$\begin{array}{c}
 ABCD \text{ divided by } BD \\
 \Downarrow \\
 \triangle ABD + \triangle DBC \\
 \Downarrow \text{ I.42 and I.44} \\
 \text{two adjacent parallelograms} \\
 \Downarrow \text{ I.29, I.14, I.30, I.33} \\
 \text{one outer parallelogram} \\
 \Downarrow \text{ additivity of equality of figures} \\
 \text{outer parallelogram} = ABCD.
 \end{array}$$

The general finite case repeats the same step for each triangular component. This is the part of the classical argument that the final Lean induction captures most directly.

4 Triangulated Figures and the Representation Boundary

The proposition is the first point in Book I where the input is not naturally one triangle or one quadrilateral but an arbitrary rectilineal figure. This forces a distinction between a geometric region and a formal decomposition of that region.

The Hilbert layer already contains

```

structure HilbertTriangle
  (Geo : Geometry.Geo) where
  A : Geo.Point
  B : Geo.Point
  C : Geo.Point

abbrev HilbertTriangulatedFigure
  (Geo : Geometry.Geo) :=
  List (HilbertTriangle Geo)

```

and a relation `HilbertSameFigure` generated by triangle splitting and triangle permutations. This is genuine infrastructure for reasoning about triangulations. It is intentionally more structural than a coordinate polygon representation.

Proposition I.45 introduces the executable term

```

def rectilinearTerm
  (L : HilbertTriangulatedFigure Geo) :
  HilbertScissorsTerm Geo :=
  (L.map
    (fun t =>
      hilbertScissorsTriangle Geo t.A t.B t.C)).sum

```

which forgets all incidence between different list entries and retains exactly what the proof needs for content: a finite formal sum of triangle terms.

This choice has two consequences.

First, it is sufficient for the theorem actually proved. The induction step uses only the identity

$$\text{Term}(T :: L) = \text{Tri}(T) + \text{Term}(L),$$

followed by addition and transitivity of equicomplementability. No operation in the proof needs to inspect a polygon boundary.

Second, the theorem should not be described as proving from first principles that every such list is a simple rectilinear region. A geometric rectilinear figure requires a compatible finite triangulation of one region, in particular the appropriate nonoverlap and coverage conditions. The current Lean theorem does not carry those conditions as hypotheses. When L comes from a genuine triangulation, the theorem formalizes Euclid I.45. Abstractly, it proves a slightly more general finite-additive statement.

This is an important architectural checkpoint. The existing `HilbertTriangulatedFigure` plus scissors terms are sufficient for the constructive content calculus of I.45. A stronger polygon-region type may still be useful later for theorems whose hypotheses or conclusions depend on the boundary, interior, overlap, containment, or equality of geometric regions. I.45 itself does not force that extension.

5 Proof Architecture of the Reconstruction

The 2300-line Lean file is best understood as eight conceptual stages rather than as a sequence of low-level incidence manipulations.

5.1 Step 1: turn a triangle list into a scissors term

The small definitions

```
rectilinearTerm
rectilinearTerm_cons
rectilinearTerm_nil
```

form the additive interface between the list representation and `HilbertScissorsTerm`. They isolate the only list algebra used by the final induction.

This stage is purely representational. It introduces no new geometric relation and no content axiom.

5.2 Step 2: prove the geometry of pasting adjacent parallelograms

Suppose $ABCD$ and $BCML$ are parallelograms and the strict orders are

$$D - C - M, \quad A - B - L.$$

The helpers

```
i45_outer_first_parallel
i45_outer_second_parallel
i45_paste_parallelgrams_geometry
```

prove that the outer quadrilateral $ALMD$ is a parallelogram.

The first outer parallel is obtained mostly by transporting parallelism along the two collinear produced sides. The second is subtler: the proof has two lines parallel to the common carrier and must show that the two candidate outer carriers are distinct before invoking Euclidean transitivity/uniqueness of parallels. This is one of the places where the formal proof uses Group IV in a way hidden by the ordinary picture.

Figure 2 records exactly this geometric paste, and nothing from the later scissors refinement.

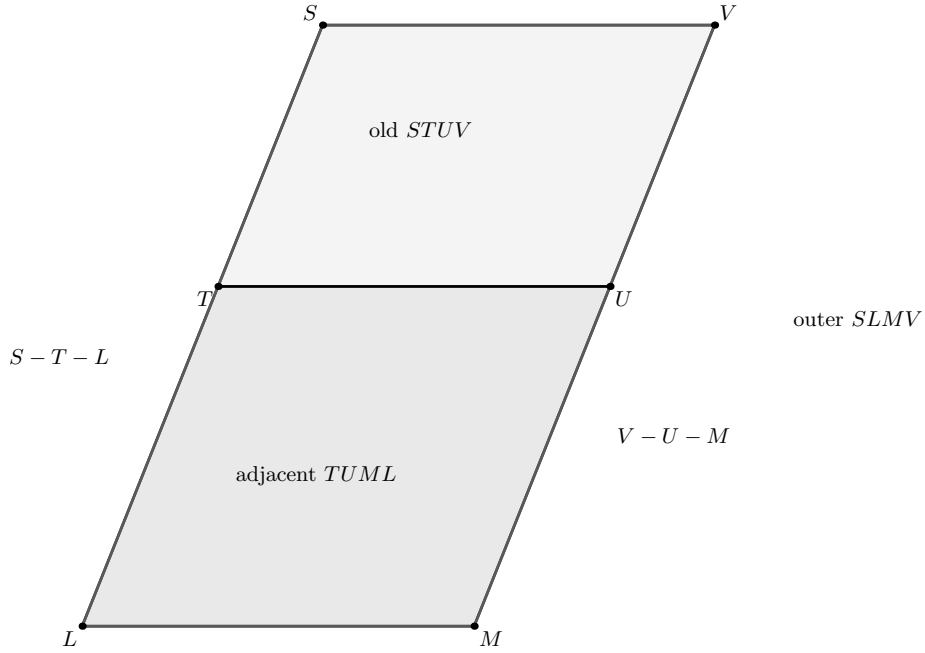


Figure 2: The formal paste configuration. The old parallelogram is $STUV$, the adjacent one is $TUML$, and the strict orders $S - T - L$ and $V - U - M$ place the second piece externally. The outer boundary is the parallelogram $SLMV$. No crossing point is shown here: its existence is a separate later theorem used only for the scissors identity.

5.3 Step 3: refine the paste into a scissors identity

The geometric union of two adjacent parallelograms is not automatically an identity of the project's multiset terms. The proof therefore constructs an internal crossing point.

The helper

```
i45_paste_crossing
```

returns a point X satisfying

$$A - X - M, \quad B - X - C.$$

Its existence is obtained by side-of-line information and a forced Pasch argument. Parallel/order transport is then used to identify the second betweenness relation.

With this witness,

```
i45_paste_parallelgrams_content_of_crossing
```

splits the relevant triangles and uses the generic diagonal-flip theorem

```
crossing_quadrilateral_two_triangulations
```

Explicit multiset associativity and commutativity then normalize the pieces. The packaged result

```
i45_paste_parallelgrams_content
```

is the formal identity

$$\text{Par}(ALMD) \sim \text{Par}(ABCD) + \text{Par}(BCML).$$

Before packaging the equicomplementability step, it is useful to make the actual scissors geometry visible. Figure 3 separates the three states consumed by the proof: the outer triangulation, the crossing witness X , and the diagonal flip inside the central quadrilateral. In the notation of the helper these relations are

$$A - X - M, \quad B - X - C.$$

They permit the replacement

$$\triangle ABM + \triangle AMC \sim \triangle ABC + \triangle BCM,$$

while the two remaining triangles are left unchanged.

(a) split the outer parallelogram

(b) crossing witness

(c) regroup as the two inner pieces

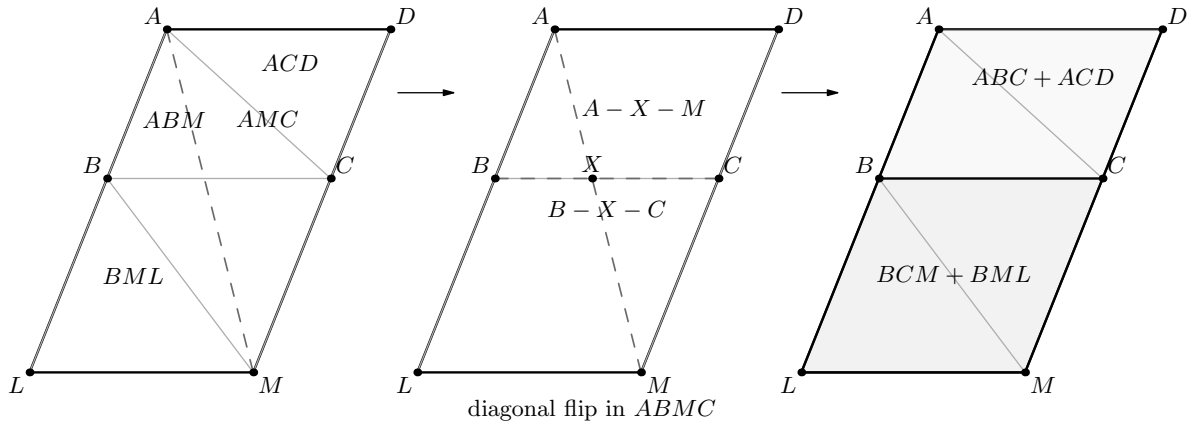


Figure 3: The scissors proof of the paste identity. First the outer parallelogram is split into the four triangle terms ABM , BML , AMC , and ACD . The point X with $A - X - M$ and $B - X - C$ certifies the two triangulations of the central quadrilateral $ABMC$. After the diagonal flip, the pieces regroup as $(ABC + ACD) + (BCM + BML)$, exactly the two inner parallelogram terms.

The companion

```
i45_paste_equicomplementable
```

combines that scissors equality with an equicomplementability certificate for the right-hand piece. This is the additive content engine used by the induction step.

5.4 Step 4: preserve the prescribed angle

The helper

```
i45_paste_angle
```

proves that pasting does not change the prescribed angle carried by the first parallelogram.

For the outer parallelogram $ALMD$, opposite angles give

$$\angle ALM \cong \angle MDA.$$

The order $D - C - M$ identifies the rays DC and DM , so

$$\angle MDA = \angle CDA.$$

The old parallelogram $ABCD$ gives

$$\angle ABC \cong \angle CDA.$$

Transitivity therefore yields

$$\angle ALM \cong \angle ABC.$$

This argument uses only the developed parallelogram-angle theorem and SameRay angle transport; it introduces no new angle principle.

5.5 Step 5: reduce the orientation problem to one strict order

The main obstruction to using public I.44 directly is orientation. Public I.44 returns a parallelogram on a prescribed segment, but it does not expose which side of the segment contains the new parallelogram. I.45 needs the new piece to be external to the old one.

The first reduction is

```
i45_other_side_order
```

which shows that for parallelograms $STUV$ and $TUML$,

$$V - U - M \implies S - T - L.$$

The proof uses TU as a separator. Parallel endpoints give SameSide data for V, S and for M, L . Since $V - U - M$, points V and M are on opposite sides of the carrier TU . OppositeSide transport therefore puts S and L on opposite sides. Euclidean uniqueness of the parallel identifies the carrier SL through T , and incidence uniqueness identifies the crossing point with T . Thus only one strict order must be controlled by the construction.

The second reduction is

```
i45_extension_angle
```


If $STUV$ is a parallelogram and $T - U - E$, then

$$\angle EUV \cong \angle STU.$$

The proof explicitly constructs the side-of-line configuration for the transversal, invokes the developed alternate-angle theorem for parallel lines, and replaces interior points by the corresponding SameRay endpoints. This lemma makes it possible to orient the copied I.44 shape using the old ray UV itself.

5.6 Step 6: construct the new triangle piece on the correct side

The central new helper is

```
i45_oriented_triangle_placement
```

Starting from $STUV$ and a nondegenerate triangle PQR , it constructs points E, F, G such that

$$T - U - E, \quad UFG \text{ is a parallelogram}, \quad G \text{ lies on the ray } UV,$$

and

$$\triangle PQR \approx UFG.$$

The proof first uses I.42 to obtain a source parallelogram with the correct triangle content and the angle STU . It then extends TU beyond U and lays off one copied side along that extension. The other copied side is laid off directly on the old ray UV . The lemma `i45_extension_angle` identifies the included angle, and the existing I.44 helper `i44_placed_parallelogram_content` certifies the piecewise scissors content.

Figure 4 shows this stage without imposing an order between V and G that Lean does not prove. The first panel fixes the old ray UV ; the second shows the copied parallelogram and places G on that ray, but omits V from the second panel. Thus the diagram records precisely a SameRay fact rather than an accidental betweenness relation.

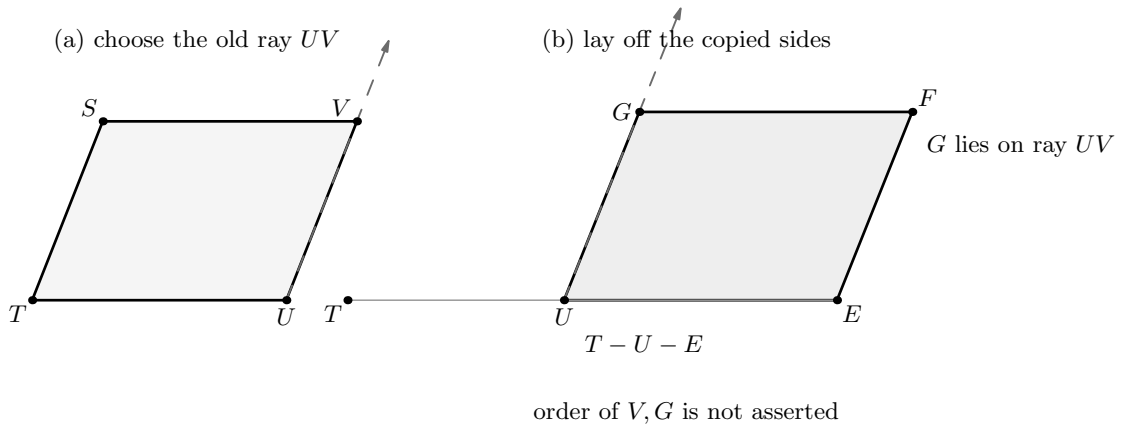


Figure 4: The orientation step before the ordered I.44 construction. Panel (a) fixes the selected ray UV of the old parallelogram. Panel (b) extends TU to E , giving $T - U - E$, and lays off G on the same ray from U as V before completing UFG . No order between V and G is asserted; the formal conclusion is only $\text{SameRay}(U; V, G)$.

The next helper

```
i45_positioned_content_ordered
```

is a deliberately stronger local wrapper over the internal I.44 construction. The public helper `i44_positioned_content` returns the target parallelogram and its content equivalence but discards the intermediate order fact $G - B - M$. I.45 needs that fact, so it reuses the same I.44 construction of H, K, L, M and retains

$$G - B - M$$

before hiding the internal witnesses.

This is a representation/API issue, not a new geometric principle. No part of I.44 is assumed again. The local theorem replays already proved I.44 helpers and exposes one conclusion needed by the stronger downstream orientation invariant.

Combining these two helpers gives

```
i45_adjacent_triangle_parallelogram
```

with the clean output

$$TUML \text{ parallelogram, } \quad V - U - M, \quad \triangle PQR \approx TUML.$$

This is the exact object required by the paste lemmas.

Figure 5 isolates the order bridge hidden inside this composition. The ordered internal I.44 construction retains $G - U - M$; the previous step supplies $\text{SameRay}(U; V, G)$. Betweenness transport then replaces G by the old boundary point V and yields exactly $V - U - M$.

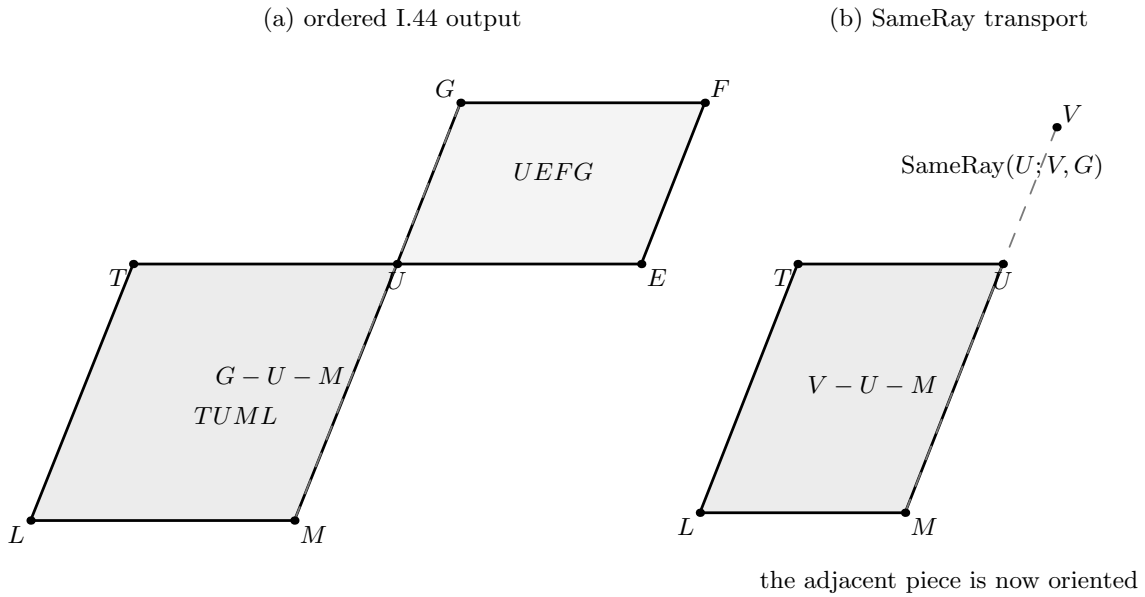


Figure 5: Panel (a) shows the extra strict order retained by the local I.44 wrapper: $G - U - M$, together with the source parallelogram $UEFG$ and the target parallelogram $TUML$. Panel (b) uses the previously established SameRay relation between V and G to transport that order to $V - U - M$. The relative order of V and G on their common ray is never needed or asserted.

5.7 Step 7: package one induction extension

The theorem

is the mathematical induction step.

Given an old parallelogram $STUV$ whose angle STU is congruent to the prescribed angle and a new triangle PQR , it first constructs the adjacent parallelogram $TUML$. The order theorem supplies $S - T - L$. The paste theorem produces the outer parallelogram $SLMV$. The angle helper preserves the prescribed angle, while the content helper proves

$$\text{Par}(SLMV) \approx \triangle PQR + \text{Par}(STUV).$$

This theorem was initially isolated as a provisional local axiom while the required orientation and paste geometry were being developed. In the final file it is a derived theorem. Eliminating that provisional assumption is the main foundational closure of I.45.

5.8 Step 8: induction on the triangle list

The public proof finally becomes short.

If the list contains one triangle, I.42 gives the required parallelogram directly. If the list contains a head triangle and a nonempty tail, the induction hypothesis gives a parallelogram for the tail and `i45_extend_parallelogram` adjoins the head triangle. The only remaining content step is `euclidcomplementable_add` followed by transitivity.

The actual Lean DAG is therefore

$$\begin{array}{c}
 L = [T_1, \dots, T_n] \\
 \Downarrow \\
 \text{rectilinearTerm}(L) = \sum_i \text{Tri}(T_i) \\
 \Downarrow \\
 \text{base: I.42 for one triangle} \\
 \Downarrow \\
 \text{oriented triangle placement on the current side} \\
 \Downarrow \\
 \text{ordered I.44 positioned construction} \\
 \Downarrow \\
 \text{adjacent parallelogram with } V - U - M \\
 \Downarrow \\
 \text{recover } S - T - L \\
 \Downarrow \\
 \text{paste geometry + scissors + angle} \\
 \Downarrow \\
 \text{i45_extend_parallelogram} \\
 \Downarrow \\
 \text{list induction} \\
 \Downarrow \\
 \text{euclid_proposition_45.}
 \end{array}$$

6 Lean Formalization

The current file imports only the immediately preceding proposition:

```
import CGJteamLab.Proposition44
```

All earlier dependencies therefore enter through the established module DAG.

6.1 The rectilinear term

The executable content of a triangulated input is

```
def rectilinearTerm
  (L : HilbertTriangulatedFigure Geo) :
  HilbertScissorsTerm Geo :=
  (L.map
    (fun t =>
      hilbertScissorsTriangle Geo t.A t.B t.C)).sum
```

The two normalization lemmas for cons and nil are used directly by the final induction.

6.2 The adjacent-piece theorem

The decisive geometric/content constructor has the following shape, written in the documentation's ASCII convention:

```
theorem i45_adjacent_triangle_parallelogram
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]
  (S T U V P Q R : Geo.Point)
  (hLeft : IsParallelogram Geo S T U V)
  (hPQR : Not (Collinear Geo P Q R)) :
  exists M L : Geo.Point,
    IsParallelogram Geo T U M L /\
    Geo.Between V U M /\
    HilbertScissorsEquicomplementable Geo
      (hilbertScissorsTriangle Geo P Q R)
      (hilbertParallelogramTerm Geo T U M L)
```

The order clause is part of the public output of this helper because the paste step genuinely depends on orientation.

6.3 The induction-step theorem

The formerly provisional construction is now a theorem:

```
theorem i45_extend_parallelogram
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]
  (S T U V P Q R X Y Z : Geo.Point)
  (hParallelogram : IsParallelogram Geo S T U V)
  (hAngle : Geo.AngleCongruent S T U X Y Z)
  (hPQR : Not (Collinear Geo P Q R)) :
  exists T' U' V' : Geo.Point,
    IsParallelogram Geo S T' U' V' /\
    Geo.AngleCongruent S T' U' X Y Z /\
```

```
HilbertScissorsEquicomplementable Geo
(hilbertParallelogramTerm Geo S T' U' V')
(hilbertScissorsTriangle Geo P Q R +
 hilbertParallelogramTerm Geo S T U V)
```

This is the clean mathematical interface between the large local geometry and the final list induction.

6.4 The public proposition

The public theorem is

```
theorem euclid_proposition_45
[HilbertIncidence Geo]
[HilbertEuclideanPlane Geo]
(X Y Z : Geo.Point)
(hXYZ : Not (Collinear Geo X Y Z))
(L : HilbertTriangulatedFigure Geo)
(hNonempty : L != [])
(hTriangles :
  forall t in L,
    Not (Collinear Geo t.A t.B t.C)) :
exists S T U V : Geo.Point,
  IsParallelogram Geo S T U V /\
  Geo.AngleCongruent S T U X Y Z /\
  HilbertScissorsEquicomplementable Geo
  (rectilinealTerm Geo L)
  (hilbertParallelogramTerm Geo S T U V)
```

The displayed code is documentation-friendly ASCII. The compiled source uses the corresponding Lean quantifier and membership notation.

The theorem requires no compatibility condition between distinct triangles in L . This is intentional in the current statement and is the reason the representation discussion above is part of the formal result rather than a purely historical aside.

7 Relationship with Earlier Propositions

7.1 Proposition I.42

I.42 is the base case of the induction and also the shape/content source inside

```
i45_oriented_triangle_placement
```

It already knows how to turn one nondegenerate triangle into a parallelogram with a prescribed angle. I.45 does not reprove that content theorem.

The new task is to place the I.42 shape in a controlled orientation relative to an already constructed parallelogram.

7.2 Proposition I.44

I.44 is the principal direct dependency of the induction step. Its internal rigid-placement certificate

```
i44_placed_parallelogram_content
```

is used to certify the oriented auxiliary copy. Its ordered construction of H, K, L, M is reused by `i45_positioned_content_ordered` so that I.45 can retain the strict order $G - B - M$.

This is a useful API lesson. Public I.44 intentionally exposes only the parallelogram, its prescribed angle, and its content. I.45 needs a stronger orientation invariant for adjacency. Rather than add an axiom or alter the meaning of public I.44, the local wrapper reuses the proved internal construction and exposes the one extra order fact downstream.

7.3 Proposition I.43

I.43 does not appear explicitly in the final induction theorem, but it remains inside the positioned I.44 construction. The content equivalence between the placed auxiliary parallelogram and the target parallelogram on the prescribed side is still the I.43 complement theorem, transported through the I.44 representation bridge.

Thus the content DAG is genuinely cumulative:

$$\text{I.42} \longrightarrow \text{I.44 (using I.43)} \longrightarrow \text{I.45}.$$

7.4 The Euclidean parallel block

Classically I.45 invokes I.29, I.30, and I.33 to recognize the final parallelogram. The Lean proof consumes the corresponding Hilbert-level parallel infrastructure rather than replaying those proposition numbers literally.

The important operations are transport of parallelism along collinear carriers, transitivity of parallels, and uniqueness of a parallel through a point. These are exactly the operations used in the paste geometry and in the proof that the second strict order follows from the first.

7.5 The content block I.35–I.41

These propositions are present transitively through I.42, I.43, and I.44. I.45 itself does not use a converse-content theorem and never infers geometry from equality of content. It only composes forward content certificates. Consequently it does not require the proper-part/positivity machinery that marks the difficult converse boundary at I.39.

8 Hilbert and Book Zero Components Used

8.1 Betweenness and SameRay

Strict betweenness is essential to adjacency. The two decisive relations for one paste are

$$S - T - L, \quad V - U - M.$$

They say more than collinearity: they determine which side of the common segment contains the new parallelogram.

SameRay transport is used at two levels. First, segment construction places copied side lengths on chosen rays. Second, angle identities replace an auxiliary point by another point on the same ray without changing the angle. The oriented placement depends critically on the fact that G is not merely collinear with U, V , but lies on the same ray from U as V .

8.2 SameSide, OppositeSide, and plane separation

The theorem `i45_other_side_order` is a concentrated use of Hilbert plane separation. Parallel endpoints supply SameSide facts, one strict betweenness supplies an OppositeSide witness, and OppositeSide transport propagates the orientation to the opposite boundary.

The crossing theorem also uses side-of-line data. The common side carrier is used as a separator, and the proof rules out one segment-line intersection before applying a forced Pasch argument. These are exactly the kinds of order facts that are visible in Euclid’s diagram but must be named in Lean.

8.3 Pasch and crossing

The scissors proof requires an actual point X lying on both the outer diagonal-like segment and the common side. The helper `i45_paste_crossing` obtains it from the order/separation machinery rather than assuming the obvious visual crossing.

The conclusion is stronger than bare line intersection:

$$A - X - M, \quad B - X - C.$$

These strict relations are what allow the triangle-splitting constructors in `HilbertScissorsEq` to be invoked.

8.4 Euclidean uniqueness of parallels

The proposition is Euclidean. The type-class context

```
[HilbertIncidence Geo]
[HilbertEuclideanPlane Geo]
```

appears throughout the major local helpers.

Group IV enters when the proof identifies carriers by uniqueness/transitivity of parallels. It is used in the outer-parallelogram geometry, in the orientation theorem that recovers $S - T - L$, and inside the I.44 machinery used for positioned content. This is the formal counterpart of the specifically Euclidean parallel reasoning in the classical I.45 construction.

8.5 Congruence and angle construction

The oriented copy uses segment construction on prescribed rays and the existing congruence theory of parallelograms. The initial source parallelogram comes from I.42. No new SAS, SSS, or angle-construction theorem is introduced in I.45; those results enter through the established interface and through I.44.

8.6 Scissors operations

The content layer uses the existing operations

```
HilbertScissorsEq.split
HilbertScissorsEq.add
HilbertScissorsEq.trans
equicomplementable_of_scissorsEq
equicomplementable_refl
equicomplementable_symm
equicomplementable_trans
equicomplementable_add
crossing_quadrilateral_two_triangulations
```

No subtraction, cancellation, positivity, or scalar area is required. The entire direction is constructive and additive.

9 Formalization Notes

9.1 Geometry

The geometric induction step requires substantially more than the phrase “apply the next parallelogram adjacent to the first.” The formal proof must establish:

- a produced point E with $T - U - E$;
- a copied point G on the selected ray UV ;
- noncollinearity needed to complete the auxiliary parallelogram;
- the ordered I.44 output $G - U - M$ and its transport to $V - U - M$;
- the opposite boundary order $S - T - L$;
- both outer pairs of opposite sides parallel;
- an actual crossing point with two strict betweenness relations for the scissors refinement;
- preservation of the prescribed angle after the paste.

None of these facts is inferred from a drawing. They are proved by the existing Hilbert order, separation, incidence, congruence, and Euclidean parallel theory.

9.2 Content

The content proof is forward and additive.

The new triangle is first shown equicomplementable with the oriented auxiliary parallelogram. The ordered I.44 construction transfers that content to the adjacent parallelogram on the common side. The paste theorem gives a direct scissors equality between the outer parallelogram and the formal sum of the two inner parallelograms. Finally `equicomplementable_add` and transitivity combine the previous parallelogram with the new triangle.

No equal-content hypothesis is ever used to deduce a line, angle, side, or order relation. Therefore no de Zolt/proper-part principle is needed. I.45 also does not use numerical area, multiplication of lengths, or cancellation of an area factor.

9.3 Representation

There are two representation issues.

The first is the triangle-list input. `rectilinearTerm` is a sum of triangle terms, and the public theorem only requires each listed triangle to be nondegenerate. It does not prove that the list is a valid nonoverlapping triangulation of one polygonal region. This should remain explicit in any reuse of I.45.

The second is the orientation information hidden by public I.44. The theorem

```
i45_positioned_content_ordered
```

duplicates no axiom; it simply replays the proved I.44 construction while retaining the intermediate order $G - B - M$. This is a local API strengthening needed to compose constructions. A future refactoring could expose such oriented placement as a shared theorem, but the present I.45 proof does not require changing the public I.44 statement.

9.4 Axiomatic status

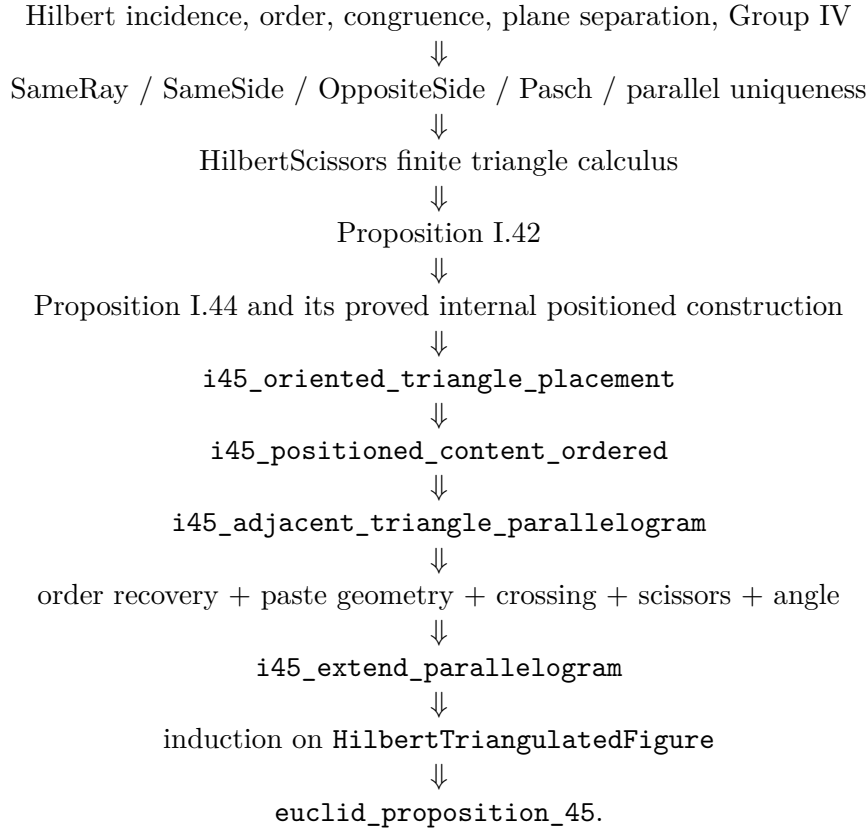
Proposition I.45 is Euclidean. Hilbert Group IV is used through the developed uniqueness/transitivity theory of parallels. The proposition also uses the incidence, order, plane-separation, congruence, and angle-construction layers inherited from the Hilbert interface and from I.44.

No continuity axiom is used by I.45. No Archimedean axiom is used. No new content axiom is introduced. The conclusion is equicomplementability, not a stronger universal equidecomposability claim.

The current source contains no active `axiom`, `sorry`, or `admit` declaration for I.45. The former provisional `i45_extend_parallelogram` assumption has been replaced by the derived theorem documented above.

10 Dependency Summary

The formal dependency chain is



The classical and formal DAGs can be compared succinctly as

Euclid: triangulate $\rightarrow$ I.42/I.44 pieces $\rightarrow$ parallel paste $\rightarrow$ add equality,
 Lean: triangle list $\rightarrow$ oriented I.44 piece $\rightarrow$ strict-order recovery
 $\rightarrow$ scissors paste $\rightarrow$ induction.

New geometric axiom:	no
New content axiom:	no
New proposition-local helpers:	yes
New reusable helper promoted to Interface/Scissors:	no
New Book Zero theorem:	no
New Interface/Scissors theorem:	no
New numerical area semantics:	no
New full polygon-region semantics:	no
Continuity used:	no
Archimedean axiom used:	no
Hilbert Group IV used:	yes
Public theorem compiles:	yes
Full lake build:	yes, 1458 jobs
Pre-existing unrelated linter warning:	yes, unused hBC
sorry in Proposition45.lean:	no

Diagrammatic audit. The final documentation uses five figures, each with one geometric role. Figure 1 is historical: it records Euclid’s two-triangle I.42/I.44 composition and does not

claim to be the full Lean witness configuration. Figure 4 records the formal ray-controlled copy and deliberately avoids asserting an order between the two SameRay points. Figure 5 records the additional strict order retained from the internal I.44 construction and its transport to the old boundary ray. Figure 2 records only the geometric union of the two adjacent parallelograms. Finally, Figure 3 adds the crossing witness and the exact retriangulation required by the scissors equality. No separate diagram is used for list induction, multiset associativity/commutativity, or final angle transitivity, since those steps introduce no new geometric configuration.

11 Conclusion

Proposition I.45 closes the application-of-areas construction by showing that the triangle-level machinery can be iterated without introducing a new area axiom. The final Lean proof is not a literal transcription of Euclid’s quadrilateral diagram. Its central operation is a derived extension theorem: construct the next triangular content as an oriented adjacent parallelogram, recover the strict boundary order, paste the geometry, refine the paste in the scissors calculus, and preserve the prescribed angle.

The most delicate issue is orientation. Public I.44 intentionally hides the side on which its target parallelogram lies. I.45 solves this synthetically, not axiomatically: one copied side is placed on the produced base ray, the other on the old parallelogram ray, the ordered internal I.44 construction is reused, and plane separation converts one retained order relation into the second. Once this is done, the rest of the proof becomes a clean additive induction.

The proposition also answers the earlier representation checkpoint in a qualified way. The existing `HilbertTriangulatedFigure` and `HilbertScissorsTerm` layers are sufficient for the constructive content statement of I.45. No stronger polygon datatype is forced by this proof. At the same time, the public theorem treats a triangulated figure as a finite list of nondegenerate triangles and does not internalize nonoverlap, coverage, boundary, or simple-polygon semantics. Those notions remain a separate possible extension of the figure layer rather than a hidden claim of I.45.

With the provisional extension axiom eliminated and the full project build successful, I.45 is therefore closed as a theorem of the existing Hilbert plus scissors architecture.